IKIRERE ORBITAL LABS AFRICA

Multipurpose nanosatellites infrastructure for Africa

Software → Firmware → Hardware

One nanosatellite. Multiple missions.

PRE-SEED ROUND · MAY 2026

The Shift We Are Driving

Computers once filled entire rooms — expensive, single-purpose, and inaccessible. Today, devices smaller than a fingernail run multiple functions and are far more powerful. **That shift was driven by compression. Space has not gone through that transition yet.**

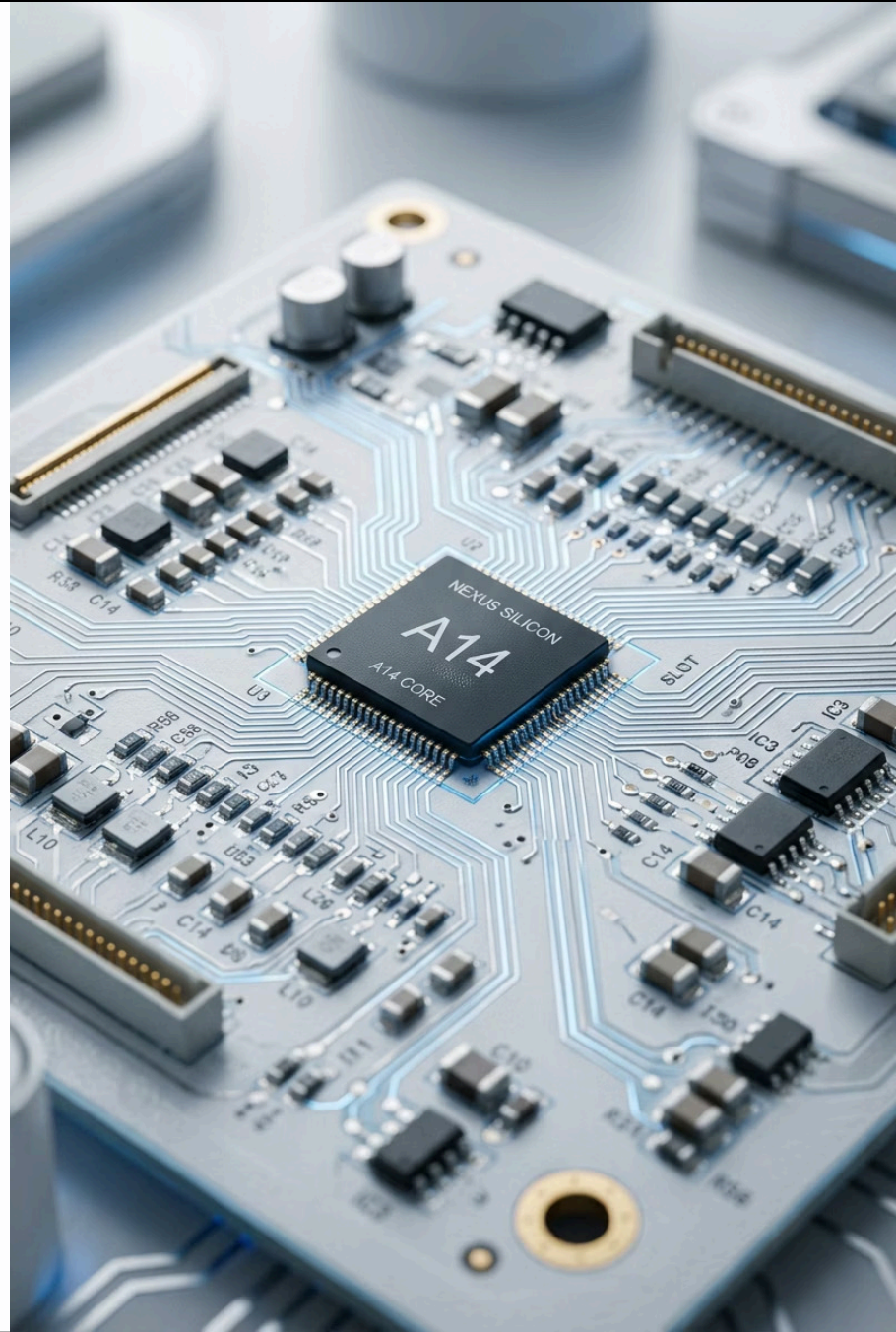
Satellites Today

- One big satellite = one mission
- Multiple satellites required
- High cost per capability
- Fragmented systems
- Long deployment cycles

IOLA's Vision

- One nanosatellite = multiple missions
- Fewer satellites required
- Lower cost per function
- Integrated system design
- Faster iteration through software

We are not adding more satellites. We are redesigning what a satellite is.



Space Has Changed. Africa's $25B space economy Has Not Caught Up Yet.

95%

Launch Cost Drop

CubeSats can now be launched for ~$6,000

$2B+

Data Demand by 2030

Satellite infrastructure is now a real asset class

17%

World Population

Africa's share — yet less than 1% of operational satellites

14

National Space Agencies

Yet Africa depends on external infrastructure

The constraint is not ambition. It is infrastructure and access.

University Labs

- Cannot afford $50K/year tools
- Cannot simulate or validate systems

Small Operators

- No coordination layer
- Manual tracking
- High operational risk

⚠ This is not just a tools problem. It is an architecture problem. Satellites are still single-purpose.

IOLA — What We Are Building

Hardware. Firmware. Ground Software. In that order of ambition, in reverse order of execution.

1 — Ground Software Stack

Now → 2027

- Simulation & real orbital mechanics (SGP4)
- Constellation design & coordination

2 — Onboard Firmware

2027 → 2028

- Mission execution & power management
- Communication scheduling & autonomous control

3 — Multipurpose Nanosatellites

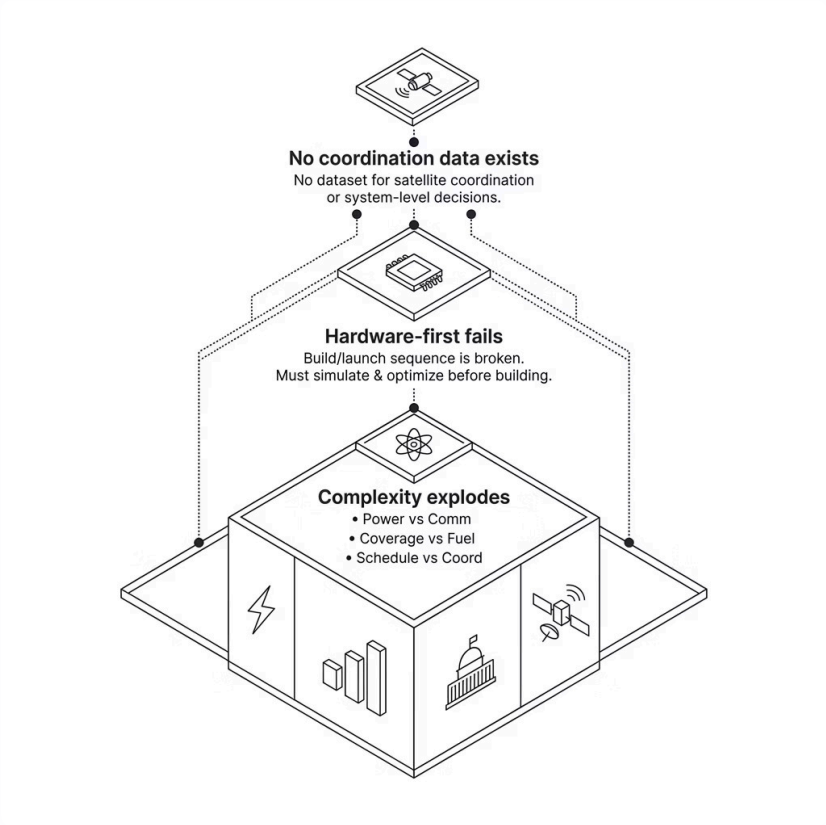
2028 → 2031

- Climate, Agriculture, Connectivity
- Logistics & Earth observation

One nanosatellite handles climate, agriculture, connectivity, logistics, and Earth observation — perfect and useful across all, reducing satellite count. Software is how we design and validate before building.

Why This Hasn't Been Done

Multipurpose nanosatellites are a different class of system. The barriers are real — and we are built to overcome them.



Complexity Explodes

- Power vs communication
- Coverage vs fuel
- Scheduling vs coordination

Hardware-First Fails

- Build → launch → validate does not work
- We simulate → optimize → then build

No Coordination Data Exists

- No dataset for real satellite coordination
- No dataset for maneuver decisions
- No dataset for system-level optimization

We generate it.

Product — Command Center + IkirereMesh

ikirere.com/sandbox — Multipurpose satellites require this coordination layer to function.

Command Center

- Simulates constellations
- Real-time orbital tracking
- Telemetry modeling

IkirereMesh

- Real-time coordination
- System optimization
- Autonomous decisions

Every Maneuver Generates Data

The system learns from every decision made in orbit

The System Improves Continuously

Coordination intelligence compounds over time

The Moat Compounds

Proprietary dataset no competitor can replicate

Demo — Command Center

ikirere.com/sandbox

Experience our live system for designing and operating satellite constellations before deployment, validating missions in a realistic simulated environment.

What it does

- Spawn constellations on a real 3D globe
- Set altitude, inclination, speed, and coverage
- Simulate multi-constellation environments

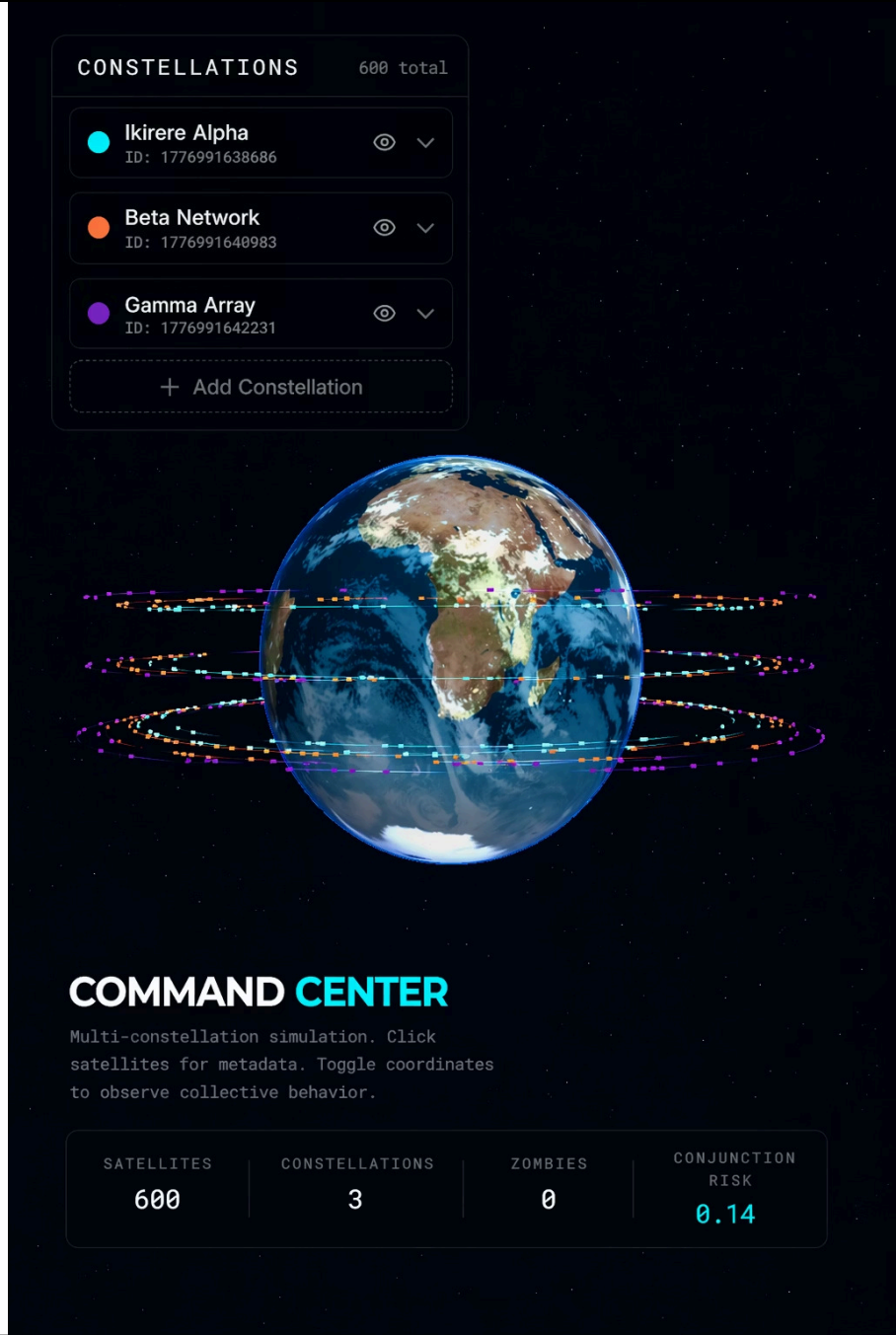
Real-time Telemetry

- Battery, solar input, fuel levels
- Thermal conditions, signal strength
- Detailed orbital dynamics

IkirereMesh in Action

- Activate coordination engine
- Watch satellites self-organize
- Reduce conjunction risk in real time

By Q3 2026, every object in the simulation maps to a real satellite using SGP4 physics, ensuring unparalleled realism.



Why We Win

Our competitors optimize parts of the system. We redesign the system itself.

IOLA

Ikirere Orbital Labs Africa

- Multipurpose satellite architecture
- Simulation → coordination → hardware
- IkirereMesh RL coordination engine
- Proprietary coordination data layer
- Built for accessibility and scale

→ Purpose-built infrastructure for the next generation

Legacy Software

Ansys STK, FreeFlyer

- Built for single-purpose missions
- $50K+/year licenses
- No real-time coordination
- No adaptive intelligence

→ Static tools for a dynamic problem

Traffic & Autonomy Platforms

Kayhan Space, Neuraspace, AIKO

- Collision avoidance only
- Coordination layer only
- No satellite architecture redesign
- Limited full-system integration

→ Partial solutions, not full infrastructure

Vertical Operators

SpaceX, Spire, Open Cosmos

- Closed internal software
- No shared coordination
- Closed ecosystems
- Not tools, not platforms

→ Closed ecosystems, not open infrastructure

Business Model + Go-To-Market

Our phased approach targets key segments, building data and traction to scale.

Phase 1: University Labs (Q3–Q4 2026)

Target: African university labs and research institutions

Channels: Deep Learning Indaba network, Google + Nvidia programs, direct outreach

Goal: 3–5 pilot customers, initial product validation, early coordination data generation

Phase 2: Small Satellite Operators (Q4 2026 – Q2 2027)

Target: Operators on SpaceX rideshares, emerging African space companies

Channels: SpaceX Transporter ecosystem, satellite operator communities, direct sales

Goal: 5–10 paying customers, ~$250K ARR, early usage-based revenue

Phase 3: Enterprise (Q2 2027 Onward)

Target: National space agencies (Kenya, South Africa, Rwanda, others)

Channels: Government procurement, strategic partnerships, aerospace networks

Goal: 2–3 anchor contracts, $750K+ ACV

Why this works

- Start with low-friction simulation for rapid adoption.
- Continuously improve IkirereMesh with real-world insights.
- Generate invaluable coordination data with each engagement.
- Convert validated solutions into high-value enterprise contracts.

Each phase strategically feeds into the next, creating a compounding advantage.

Business Model + Go-To-Market

Revenue Streams

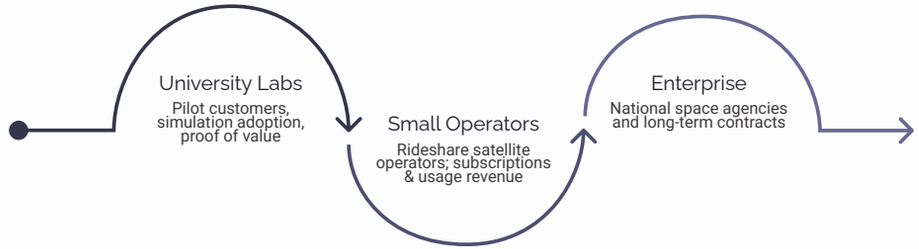
- Subscription
- Usage-based ($0.10 per decision)
- Enterprise contracts
- Future: data + infrastructure

Distribution

- Direct outreach
- Research networks, Indaba and universities
- ESA and aerospace connections

Expansion Path

Simulation → coordination → enterprise → API access → multi-constellation coordination → hardware integration

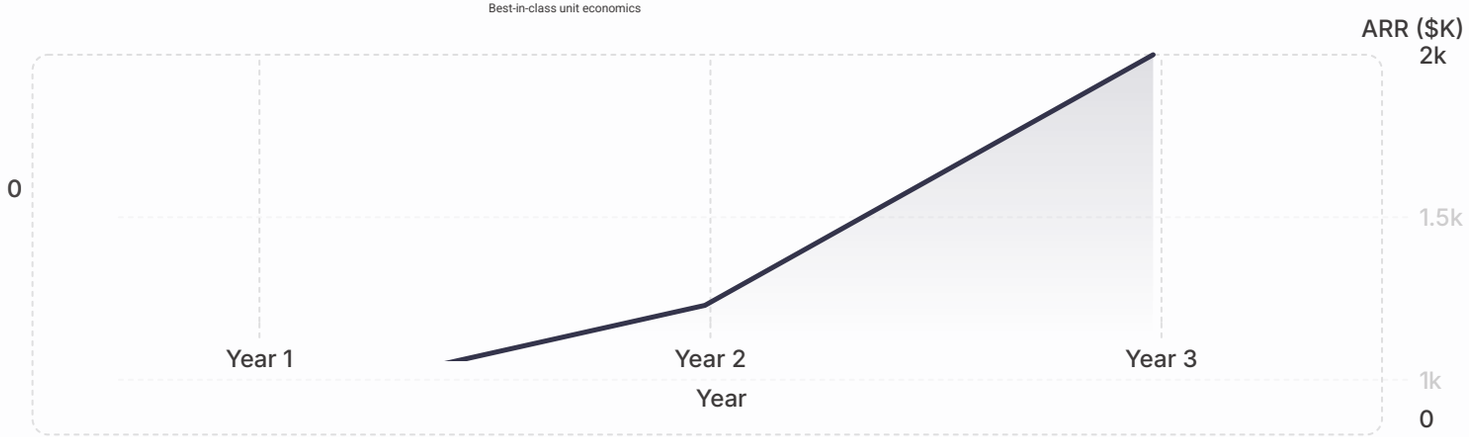


$15K
CAC
Customer acquisition cost

10:1
LTV:CAC
Best-in-class unit economics

$150K
LTV
Lifetime value per customer

75%
Margin
4-month payback period



Traction

From simulation to recognition — IOLA is already building momentum before the first satellite launches.

Deep Learning Indaba 2025

Winner at Africa's largest AI conference — validating our technical approach and continental relevance.

Google + Nvidia Inception

Compute credits and technical support from two of the world's leading technology companies.

Live Demo

1,300+ satellites simulated in real time — proving the platform works at scale today.

ESA Kick-starts

Up to €75K non-dilutive grant in progress — institutional validation from the European Space Agency.

First step: simulation. End goal: satellites in orbit.

Team

Software + Engineering + Aerospace — Africa & Europe rooted, globally relevant.

Jason Quist — CEO

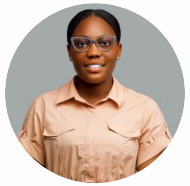
Reinforcement learning, multi-agent systems, strategy, product, and capital.

Gideon Salami — CTO

Simulation systems, orbital infrastructure, and engineering execution.

Alph Doamekpor — Strategy & Product

20+ years across the ESA ecosystem and aerospace systems alignment.



Abigail Boateng — Head of Research

Multi-agent RL, safety-critical AI, and IkirereMesh architecture.

Ignatius Balayo — ML Engineer

Research to production systems and validation pipelines.

10. We Are Raising $5M

Software-first satellite infrastructure — from **Ground** to **Orbit**.

Use of Funds

30%

Simulation Engine + Orbital Infrastructure ($1.5M)

Infrastructure and compute for the simulation engine

25%

IkirereMesh (RL Coordination Engine) ($1.25M)

Nvidia GPU access for IkirereMesh RL training

30%

Hardware Prototype & Nanosatellite Development ($1.5M)

Prototype build, lab testing, and nanosatellite development

10%

Legal & Ops ($500K)

Delaware incorporation, IP assignment, entity structure

5%

Execution Buffer ($250K)

6 months operating cushion

What This Produces in 12 Months

Validated Tracking

Real satellite tracking (SGP4) & coordination validated.

Early Wins

1 LOI, 3 research publications, and first revenue generated.

Hardware Ready

Functional hardware prototype completed and tested.

Path to $1M ARR

- 5 key customers secured
- 1 major enterprise contract
- Successful expansion into new markets

Non-Dilutive Pipeline

- European Space Agency (ESA)
- Deep Learning Indaba grants
- Google & Nvidia programs

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